

Use-Case Flow Specification

Domain & SSL Certificate Expiry Alert System

Use Case:

Send Expiry Alert

Primary Actor:

System Admin

Secondary Actor:

Security Officer

Preconditions:

The domain or endpoint is present in the monitored list (FR-003). The daily scan cycle has completed a successful TLS handshake or WHOIS lookup for that domain (FR-001, FR-002), and an expiry date has been extracted and stored.

Postconditions:

An alert record is created and logged with a timestamp, the domain name, and the days remaining until expiry. The System Admin has been notified for the applicable threshold (3, 15, or 30 days).

Main Success Scenario

1. Scheduler triggers the daily scan job.
2. System performs a TLS handshake (for certificates) or WHOIS lookup (for domain registration) against the monitored domain.
3. System calculates the number of days remaining until expiry.
4. System compares the remaining days against the configured alert thresholds (3, 15, 30 days).
5. If the remaining days matches a threshold, the system generates an alert.
6. System dispatches the alert notification to the System Admin.
7. System logs the notification event, including timestamp and threshold triggered.
8. Use case ends successfully - alert sent and logged.

Alternate Flow – 6a Alert Unacknowledged

- 6a1. The SysAdmin does not acknowledge the 3-day threshold alert within 24 hours.
- 6a2. System automatically escalates the alert to the Security Officer.
- 6a3. The escalation event is logged using the same logging path as the main flow.
- 6a4. Use case resumes at step 8.